11.10 A Ball That Comes Back Towards You

WORKSHEET

The question

STEP II 2019, Question 9

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Why is it easier to work with OP^2 than with OP? Is "OP is increasing" the same as "OP^2 is increasing"?

STAGE 1 The squared distance

Covers: Part (i), OP increasing

Ask yourself Work with $OP^2 = x^2 + y^2$ rather than OP. Write it in terms of t and simplify using $\cos^2\alpha + \sin^2\alpha = 1$.

Pause and try Differentiate with respect to t and take out a factor of t. You are left with a quadratic in t. When is it positive for every t?

STAGE 2 Steeper throws

Covers: Part (i), steeper throws

Ask yourself If $\sin\alpha > \frac{2\sqrt{2}}{3}$, the quadratic has two real roots. Use their sum and product to show both are positive. Is the smaller one before the particle lands?

STAGE 3 A moving observer

Covers: Part (ii)

Ask yourself Q is at $(-vt, 0)$, so the horizontal separation is $x + vt$. Write QP^2. How does it differ from OP^2? Repeat the discriminant argument.

What's really going on

A geometric view

OP decreases exactly when the velocity of P points "inwards", making an obtuse angle with the vector $\overrightarrow{OP}$, that is, when $\overrightarrow{OP} \cdot \mathbf{v} < 0$. For a steep throw, near the top of the flight the particle is high above and slightly in front of the thrower, moving almost horizontally away, but starting to fall; once the downward part of its velocity outweighs the outward part, relative to the direction of $\overrightarrow{OP}$, it comes closer. For shallow throws the particle is always far enough in front that its velocity never points back towards O. The critical case is when the velocity is perpendicular to $\overrightarrow{OP}$ at exactly one instant, the moment when the distance pauses in Figure 11.10.1 (in the full section online).

Monotonicity is not automatic

Many quantities that "obviously" increase do not. The distance of the Moon from the Earth oscillates; the distance between two cars on a winding road can decrease even when one is always ahead; the population of a predator can fall while its prey multiplies. Proving that a quantity is monotonic usually means showing that a derivative has a fixed sign, and the technique here (reduce to a polynomial, then check its roots or discriminant) is the standard way to do it. In analysis, monotonicity is often the key to proving convergence: a bounded monotonic sequence converges, as you saw in Chapter 2.

Try this next

1 **How close?** For $\alpha = 80°$, find the times at which OP has its local maximum and minimum, and the distances there, in units of u^2/g.

2 **On a slope.** Repeat part (i) for a ball thrown up a plane inclined at angle β to the horizontal, landing on the plane. Does the critical angle (measured from the plane) change?

3 **With air resistance.** Suppose the ball also experiences a resistance proportional to its velocity. Without calculating, would you expect the critical angle to be larger or smaller than 70.5°? Why?

4 **Throwing from a height.** If the ball is thrown from the top of a cliff and lands far below, show that OP eventually increases for any angle, and find whether there are angles for which it decreases at some stage.

Further reading

- **James Walker, *The Flying Circus of Physics*** (Wiley). Hundreds of surprising everyday physics puzzles.
- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 3, on projectiles, with several counter-intuitive problems.
- **J. M. Steele, *The Cauchy–Schwarz Master Class*** (Cambridge University Press). The art of proving that one thing is always bigger than another.

The full guided solution is in Section 11.10:
https://beyondstep.keithmcnulty.org/ch11/coming-back.html